

Amit Vyas

Columbus, OH | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

EDUCATION

M.S. Computer Science, Cleveland State University, Cleveland, OH, USA

Aug 2021 – May 2023

GPA: 3.8

B.Tech Computer Science (Minor: Big Data), Manipal Institute of Technology, Manipal, India

Jul 2015 – Jun 2019

Thesis: Helmet Detection for Safety | Advisor: Anjana S.

RESEARCH EXPERIENCE

Research Software Engineer, ICICLE AI Institute

Jan 2025 – Present

The Ohio State University, Columbus, OH

- Co-developed a GNN-based framework for multi-type food flow prediction and scenario analysis across U.S. counties; accepted for oral presentation, ACM SIGSPATIAL 2026 (Applications Track).
- Designed the architecture for the ICICLE AI Training Catalog as a documentation framework for the AI ecosystem; accepted for publication, SEHET26 Workshop (PEARC 2026) and the Journal of Computational Science Education (JOCSE).

Graduate Research Assistant, ICAS Lab

Oct 2021 – May 2022

Department of Computer Science, Cleveland State University, Cleveland, OH

- Configured an NVIDIA GPU + CARLA simulation environment and generated 5K+ multi-view KITTI-format samples to train YOLO-based 2D and geometry-informed 3D object detection models.
- Improved 3D detection accuracy via multi-camera self-attention fusion, achieving AP scores of 54.1 (Easy), 44.7 (Moderate), and 27.6 (Hard).
- Co-authored *Multi-camera 3D Object Detection via Deep Learning and Self-attention*, contributing architecture diagrams, algorithms, and the methodology section.

Undergraduate Researcher

Mar 2018 – Nov 2018

Department of Computer Science, Manipal Institute of Technology, Manipal, India

- Surveyed 10+ papers on network graph connectivity and information flow in online social networks.
- Clustered 1,000+ Twitter posts via cosine-similarity to identify topic groups and applied NLTK-based sentiment analysis to study misinformation patterns.

PROFESSIONAL EXPERIENCE

Research Software Engineer, AI Backend | The Ohio State University, Columbus, OH

Jan 2025 – Present

- Cut cached query latency 49.1x by building a Redis caching layer over a 0.6B Qwen embedding service (FastAPI, llama.cpp, Qdrant HNSW indexing) on TACC Kubernetes.
- Improved RAG response quality across open-weight models (Qwen, Llama 70B) by building a G-Eval LLM-as-judge framework with MLflow logging to score relevance, faithfulness, and coherence on every release.
- Reduced fine-tuning time 75% by orchestrating multi-GPU SLURM jobs with MLflow experiment tracking for the ICICLE Chatbook RAG assistant.
- Shipped documentation releases 4.2x faster by building a multi-agent Claude Code pipeline (6 skills, 5 parallel agents) automating validation, build, and deployment.
- Cut developer documentation lookup time 80% by building an MCP server (FastMCP, JSON-RPC 2.0) exposing 118+ AI components in Claude Code and Cursor, grounded by a Neo4j knowledge graph.

Technical Founder | Ecothos, Columbus, OH

May 2025 – Mar 2026

- Architected a multi-tenant RAG platform (pgvector, GCP Cloud Run) for document-based carbon analysis, serving 12+ active users with zero downtime.
- Cut LLM inference cost 70% and prompt size 90% via top-k context capping and prompt compression.
- Built AI workflows with Gemini 2.5 Flash + RAG for sustainability knowledge cards, chat-with-PDF insights, and a voice-to-insight pipeline for CO₂-savings metrics.
- Deployed cloud-native infrastructure on GCP Cloud Run with Docker and GitHub Actions CI/CD, achieving 99.9% uptime.
- Semi-Finalist, NY Nucleate Activator (Dec 2025) — selected from 200+ teams in climate-tech.

Software Engineer, Backend | Computech Corporation, Detroit, MI*Sep 2023 – Jan 2025*

- Saved \$100K+/yr by building an in-house data enrichment service (Python, Django, NLTK, spaCy, BeautifulSoup), replacing a vendor contract in 8 weeks.
- Architected a multi-tenant document automation backend (Celery + RabbitMQ), cutting API response time 40% and enforcing tenant isolation via RBAC and Azure AD SSO for 4,000+ users.
- Led development of 200+ OpenAPI-documented DRF REST endpoints across three Django apps for analytics dashboards and real-time alerting.
- Owned observability via Prometheus, Grafana, and Django logging middleware.

Software Engineer, Backend | Skillnet Solutions Inc., Twinsburg, OH*Jun 2023 – Aug 2023*

- Cut data retrieval time 93% via MS SQL Server index redesign, unblocking a Reebok production release.
- Shipped 25+ REST APIs (Django, Java, Spring Boot) integrated with a React front-end, 2 weeks ahead of schedule.

Data Analytics Intern | Medical Mutual of Ohio, Brooklyn, OH*May 2022 – Aug 2022*

- Improved data quality 6% by eliminating unused data, outliers, and missing values using Jira Query Language.
- Built a Groovy script to compute coefficient of variance for 105K+ records in a custom Jira chart.

Software Engineer, Full Stack | Indian Institute of Technology, Jodhpur, India*Oct 2020 – Jan 2021*

- Led a team of 3 to design and ship a remote health monitoring web application.
- Devised a stethoscope + omnidirectional microphone method for capturing heartbeat sounds, cutting patient data collection time 40%.

Software Engineer, Big Data | Accenture Solutions Pvt Ltd, Bengaluru, India*Oct 2019 – Jul 2020*

- Designed scalable Django APIs and Hadoop MapReduce pipelines processing 2M+ records.
- Built Tableau dashboards surfacing demographic insights across 150K+ sales records.

Machine Learning Engineer Intern | Unilever, Bengaluru, India*Jan 2019 – Jun 2019*

- Fine-tuned a YOLO object detection model to 92% accuracy on a custom dataset using Python and OpenCV.
- Automated the data pipeline end-to-end via crontab-scheduled scripts, cutting storage 41.6%.

TEACHING & ADVISING**Mentor, OSC STEM Institute, Ohio Supercomputer Center, Columbus, OH***Summer 2026*

- Taught GPU computing, PyTorch, Pandas, and NumPy to high school students in a two-week residential STEM Institute, introducing hands-on AI/ML and data science workflows on HPC infrastructure.

Graduate Teaching Assistant, Advanced Operating Systems*Sep 2022 – May 2023**Department of Computer Science, Cleveland State University, Cleveland, OH*

- Assisted Prof. Janche Sang, managing a class of 65+ students — facilitating discussions, delivering instructional materials, and grading.
- Debugged concurrent C/CUDA assignments covering POSIX threads, multiprocessing, MPI, and GPU kernels.

Volunteer Instructor | Teach Code for Good*Jan 2018 – Nov 2018**Manipal, KA, India*

- Taught programming fundamentals to 25+ school students (Grades 6–9) through interactive sessions on HTML and Python.

PUBLICATIONS

- Chen, J., Shen, J., Kang, H., Hong, Z., Jiang, Q., Bose, S., Zhang, Y., Leng, L., **Vyas, A.**, Mao, L., Ouyang, S., Zhou, K., & Qin, L. "AgentSpec: Understanding Embodied Agent Scaffolds Through Controlled Composition." *arXiv:2606.14674*, June 2026. arxiv.org/abs/2606.14674
- Zhang, Q., **Vyas, A.**, Guzman, C., Miller, M., Morales, A., Xie, Y., & Gao, S. "Multi-Type Food Flow Prediction and Scenario Analysis between U.S. Counties using Graph Neural Networks." *ACM SIGSPATIAL 2026*, Applications Track. Accepted for publication and oral presentation.
- **Vyas, A.** "A Documentation Catalog for the AI Ecosystem: Design of the ICICLE AI Training Catalog." Presented at the *SEHET26 Workshop*, PEARC 2026, Minneapolis, MN, July 27, 2026. Accepted for publication in the *Journal of Computational Science Education (JOCSE)*.
- "Multi-Camera 3D Object Detection for Autonomous Driving Using Deep Learning and Self-Attention Mechanism", *IEEE Access*, June 2, 2023. A. Hazarika, **A. Vyas**, M. Rahmati, Y. Wang.

- “Information Abuse in Twitter and Online Social Networks: A Survey”, *International Conference on Applications and Techniques in Information Security*, November 9, 2018. M.J. Akshay, N.A. Mohammed, **A. Vyas**, P.B. Shony, S.K. Correa.

BOOK

- “Artificial Intelligence in Healthcare: Advantages and Disadvantages”, ISBN: 978-81-19534-23-4, August 21, 2023. **Vyas, Amit**; Dixit, Varun; Kaur, Davinderjit; Byeon, Dr. Haewon.

OTHER PROFESSIONAL ACTIVITIES

- IEEE Access reviewer; peer-reviewed 40+ papers.
- Reviewer, ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems (MemAgents), Apr 27, 2026, Rio de Janeiro, Brazil; reviewed 3 papers on memory architectures for LLM-based agents.
- Reviewer, COLM 2026 Workshop on Efficient Reasoning (2nd Edition), Conference on Language Modeling, 2026.
- Reviewer, World Modeling Workshop (WMW) 2026, Mila – Quebec AI Institute, Montréal, Canada, Feb 4–6, 2026.
- Program Committee, HPDC 2026 RSE Lightning Talks, ACM International Symposium on High-Performance Parallel and Distributed Computing, Case Western Reserve University, Cleveland, OH, Jul 13–16, 2026.
- Judge and Mentor, Make OHIO 2025, Fontana Labs, The Ohio State University, Columbus, OH, Mar 28–30, 2025.
- Participated in AI Exploration Day 2025 at Hudson Middle School, showcasing AI use cases like animal object detection and classification to over 100 middle school kids, Hudson, Ohio, May 2, 2025.
- Presentation on “Using GenAI to Generate Synthetic Data and Train a Predictive AI Model” at Hudson Middle School, Hudson, Ohio, May 8, 2026; presented to over 100 students.
- Judge and Mentor, SummerCodex, Virtual, Jul 21–28, 2025.
- Talk: “Using Open Source Tools for Technical Documentation” at SummerCodex, Virtual, Jul 24, 2025.
- Talk: “Using Open Source Tools for Technical Documentation” at Pittsburgh Techfest, Tepper School of Business, Carnegie Mellon University, Pittsburgh, PA, Oct 24, 2025.
- Talk: “AI4CI4AI for Software and Workflows” at OSU AI Club Research Expo, Curl Viewpoint, The Ohio State University, Columbus, OH, Oct 30, 2025.
- Workshop: “GenAI to train Predictive AI” at High School I/O, Fontana Labs, The Ohio State University, Columbus, OH, Nov 23, 2025.

AWARDS

- ESAC Rising Star Award, College of Engineering, The Ohio State University, May 2026.
- 1st Place AIB (Acumen) at TechTatva, National Technical Festival of MIT Manipal, India, Oct 5, 2017.
- Finals of the Philips Data Science Platform Hackathon, Philips Innovation Campus, Bangalore, India, Oct 13, 2017.
- Semi-Finalist, Nucleate Activator New York (2025), selected among early-stage founders innovating in the Eco track.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C/C++

AI / ML: LLMs (GPT, Claude, Gemini, Qwen, Llama), LangChain, LlamaIndex, RAG, FAISS, Qdrant, pgvector, HuggingFace, PyTorch, YOLO, GNNs (PyTorch Geometric), Prompt Engineering, Agentic AI, MCP, HNSW/Cosine Similarity, MMR Reranking, G-Eval/LLM-as-Judge, MLflow, SLURM

Backend & APIs: FastAPI, Django, DRF, Spring Boot, Flask, OpenAPI 3.0, REST, MCP, JSON-RPC 2.0, Microservices

Infrastructure: AWS (SageMaker, Lambda, S3), GCP Cloud Run, Kubernetes, Docker, CI/CD, TACC/HPC, Prometheus, Grafana, GitHub Actions, Azure DevOps

Databases: PostgreSQL, pgvector, Qdrant, Neo4j, MongoDB, MS SQL Server, Redis, MySQL, SQLite

Data & Tools: NLTK, spaCy, Pandas, NumPy, Matplotlib, Knowledge Graphs, Celery, RabbitMQ, Unicorn, Nginx, Tableau, Docusaurus

Auth & Infra: JWT/JWKS, OAuth 2.0, SSO, RBAC, Tapis

Other: React, Bootstrap, Material-UI, Git, Jira, Postman, Linux, Agile/Scrum, Kanban